

CYBERSECURITY INTERNSHIP

WEEK 2 ASSIGNMENT REPORT

Advanced Networking, Analysis & Security Concepts

Submitted By:

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Internship Domain:

Cybersecurity

Week:

Week 2

Topics Covered:

- Ports & Services
- Networking Devices
- Network Security
- Packet Analysis
- Network Scanning
- LAN Configuration
- MITM Concepts

Tools Used:

- Wireshark
- Nmap
- Cisco Packet Tracer

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1: PORTS & SERVICES

Introduction: A network port is a logical communication endpoint used by applications and services to communicate over a network. Ports help identify which service or application should receive network traffic. TCP and UDP are two commonly used transport-layer protocols.

Port	Protocol	Service	Usage
20	TCP	FTP Data	Transfer files/data
21	TCP	FTP	File transfer control
22	TCP	SSH	Secure remote login
23	TCP	Telnet	Remote login
25	TCP	SMTP	Sending emails
53	TCP/UDP	DNS	Convert domain names
67	UDP	DHCP Server	Assigns IP addresses
68	UDP	DHCP Client	Receives IP configuration
69	UDP	TFTP	Simple file transfer
80	TCP	HTTP	Web communication
110	TCP	POP3	Receiving emails
123	UDP	NTP	Time synchronization
143	TCP	IMAP	Email access and management
161	UDP	SNMP	Networking monitoring
389	TCP/UDP	LDAP	Directory services
443	TCP	HTTPS	Secure web communication
445	TCP	SMB	File and print sharing
587	TCP	SMTP Submission	Sending/submitting emails
993	TCP	IMAPS	Secure email access
3389	TCP	RDP	Windows remote desktop

TASK 2: NETWORKING DEVICES

1. Router: A router is a networking device that connects different networks and forwards data packets from one network to another. It uses IP addresses to determine the best path for sending data.

Working: When a packet reaches a router, the router checks the destination IP address. It looks at its routing table and forwards the packet toward the appropriate network. A router can also provide functions such as Network Address Translation (NAT) and DHCP in many home and small-office networks.

Real-World Example: A home Wi-Fi router connects devices such as smartphones, laptops, and smart TVs to the Internet. It forwards traffic between the home network and the Internet.

2. Switch: A switch is a networking device used to connect multiple devices within the same Local Area Network (LAN). It mainly uses MAC addresses to forward Ethernet frames to the correct device.

Working: A switch learns the MAC addresses of connected devices and stores them in a MAC address table. When it receives a frame, it checks the destination MAC address and forwards the frame through the appropriate port.

Real-World Example: In an office, several computers and printers can be connected to one network using an Ethernet switch. This allows the devices to communicate with each other over the local network.

3. Firewall: A firewall is a security device or software that monitors and controls incoming and outgoing network traffic according to predefined security rules.

Working: A firewall examines network traffic and decides whether to allow or block it. Rules can be based on factors such as IP addresses, ports, protocols, and connection information.

Real-World Example: A company may use a firewall between its internal network and the Internet to block unauthorised connections while allowing legitimate business traffic.

Router	Switch
Connects different networks	Connects devices within a LAN
Mainly uses IP addresses	Mainly uses MAC addresses
Works primarily at the Network Layer	Primarily operates at the Data Link Layer
Determines routes between networks	Forwards frames within a network
Commonly connects a LAN to the Internet	Commonly connects PCs, printers and other LAN devices

TASK 3: NETWORK SECURITY BASICS

1. What is Network Security?

Network security is the practice of protecting computer networks, devices, and data from unauthorized access, misuse, attacks, and other security threats. It uses technologies, security rules, and procedures to maintain the confidentiality, integrity, and availability of network resources.

Main Goals of Network Security

- **Confidentiality:** Ensures that information is accessible only to authorized users.
- **Integrity:** Ensures that data is not improperly modified or altered.
- **Availability:** Ensures that network services and resources remain accessible when needed.
- **Authentication:** Verifies the identity of users or devices.
- **Access Control:** Ensures users can access only the resources they are authorized to use.

2. Man-in-the-Middle (MITM) Attack: A Man-in-the-Middle attack occurs when an attacker secretly positions themselves between two communicating parties and attempts to intercept or manipulate their communication.

How it works — Normally: User → Network → Website

In a MITM situation: User → Attacker → Website

The attacker attempts to observe or alter communication while the victim believes they are communicating directly with the intended destination. Eg. An attacker may create or compromise a network in a public location, such as a Wi-Fi hotspot. If a victim connects to an unsafe network, the attacker may attempt to intercept network communication.

Risks

- Theft of sensitive information
- Session or credential exposure
- Modification of communications
- Privacy loss

Basic Prevention

- Use HTTPS websites.
- Avoid connecting to unknown or unsecured Wi-Fi networks.
- Use a trusted VPN when appropriate.
- Keep operating systems and applications updated.
- Use strong authentication, including multi-factor authentication (MFA) where available.

3. Denial of Service (DoS) Attack: A Denial Service (DoS) attack is an attack in which an attacker attempts to make a network, server, or online service unavailable to legitimate users.

How it works — The attacker sends a very large number of requests or network traffic toward a target. The target may become overloaded and may not have enough resources to respond to legitimate users.

Normal: Users → Server → Service available

During DoS: Attacker → Large amount of traffic → Server → Service becomes unavailable

Basic Prevention

- Use firewalls and network filters.
- Monitor unusual traffic patterns.
- Use DDoS protection services where appropriate.
- Keep network devices and systems properly configured and updated.

4. Packet Sniffing: Packet sniffing is the process of capturing and analysing network packets traveling across a network. It can be used for legitimate purposes such as troubleshooting and security monitoring, but unauthorized interception can expose sensitive information.

How it works — A packet-sniffing tool observes network traffic and captures packets. The captured packets can then be analysed to understand information such as protocols, source and destination addresses, and communication patterns. For example: Device → Network Traffic → Packet Capture Tool → Analysis. Tools such as Wireshark can be used by network administrators and security professionals to analyse traffic.

Basic Prevention

- Use encrypted protocols such as HTTPS.
- Avoid sending sensitive information over insecure networks.
- Use secure Wi-Fi networks.
- Use VPNs when appropriate.
- Properly configure network access controls.

5. Importance of Network Security: Network security is important because modern organizations depend on networks to communicate, store information, and provide services. Effective security measures help protect sensitive data, prevent unauthorized access, reduce service disruption, and maintain reliable network operations.

Conclusion: DoS, and packet sniffing are important network security concepts. Understanding how these threats work at a conceptual level helps cybersecurity professionals identify risks and apply appropriate protective measures. Using encryption, secure networks, monitoring, firewalls, and proper access controls can significantly improve network security.

Task 4 — Wireshark Packet Analysis

Step 1: Objective: The objective of this practical is to capture and analyse real network traffic using Wireshark. The analysis focuses on identifying DNS requests, HTTPS traffic, and the source and destination IP addresses of captured packets.

Step 2: Tool Used: Wireshark 4.6.8 — Wireshark is a network protocol analyzer used to capture and inspect network packets. It helps cybersecurity and networking professionals understand how data travels across a network.

Step 3: Step 1: Open Wireshark: Wireshark was opened on the Mac system. The available network interfaces were displayed on the Wireshark start screen.

Step 2: Then we must select the Wi-Fi interface: The Wi-Fi: en0 interface was selected because the system was connected to the Internet through Wi-Fi.

Step 3: Start packet capture: The Wi-Fi interface was opened by double-clicking Wi-Fi: en0. Wireshark then started capturing network packets.

Step 4: Generate network traffic: While the capture was running, web browsers were used to visit websites such as Google and Wikipedia. This generated DNS and web traffic that could be observed by Wireshark.

Step 5: Stop the capture: After generating sufficient traffic, the capture was stopped using the red stop button in Wireshark.

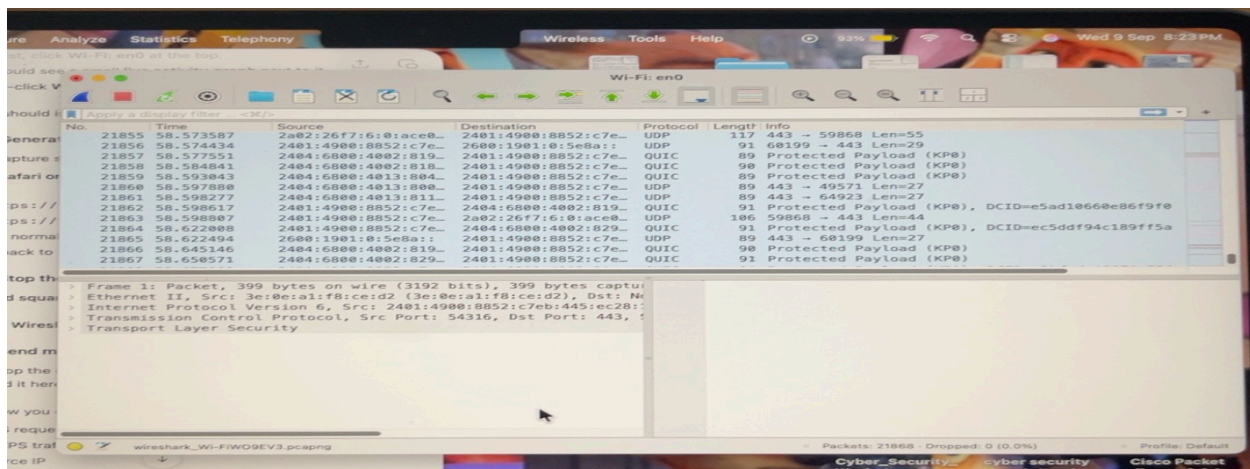
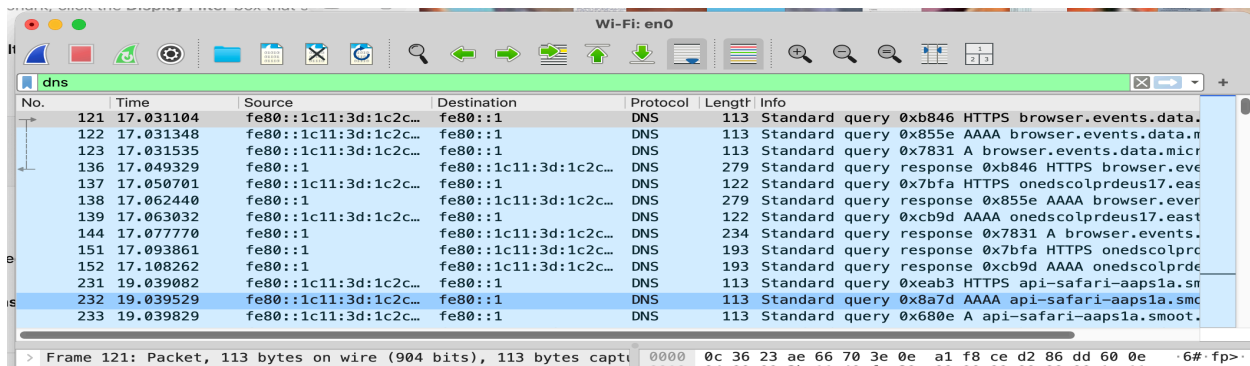


Figure 1 shows Network Packets Captured Using Wireshark

Step 4 – DNS Analysis

DNS Request Analysis: DNS (Domain Name System) is used to translate domain names into IP addresses. A DNS request is generated when a device needs to find the IP address associated with a domain name.

To identify DNS packets, the Wireshark display filter `dns` was entered in the display filter bar. The resulting packet list displayed DNS queries and responses.



The image shows a Wireshark packet capture window with the display filter set to 'dns'. The packet list shows several DNS queries and responses. The packet details pane for packet 121 is expanded, showing the DNS query structure.

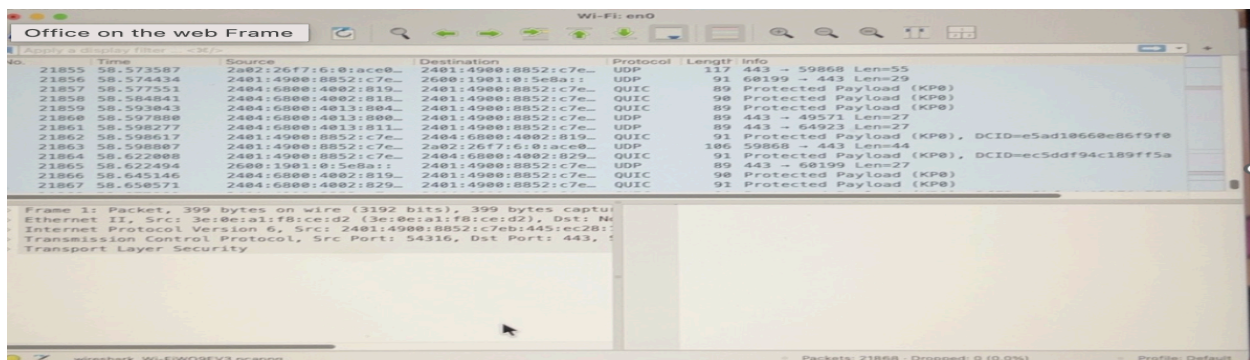
No.	Time	Source	Destination	Protocol	Length	Info
121	17.031104	fe80::1c11:3d:1c2c...	fe80::1	DNS	113	Standard query 0xb846 HTTPS browser.events.data.
122	17.031348	fe80::1c11:3d:1c2c...	fe80::1	DNS	113	Standard query 0x855e AAAA browser.events.data.m
123	17.031535	fe80::1c11:3d:1c2c...	fe80::1	DNS	113	Standard query 0x7831 A browser.events.data.micr
136	17.049329	fe80::1	fe80::1c11:3d:1c2c...	DNS	279	Standard query response 0xb846 HTTPS browser.eve
137	17.050701	fe80::1c11:3d:1c2c...	fe80::1	DNS	122	Standard query 0x7bfa HTTPS onedscolprdeus17.eas
138	17.062440	fe80::1	fe80::1c11:3d:1c2c...	DNS	279	Standard query response 0x855e AAAA browser.eve
139	17.063032	fe80::1c11:3d:1c2c...	fe80::1	DNS	122	Standard query 0xcb9d AAAA onedscolprdeus17.eas
144	17.077770	fe80::1	fe80::1c11:3d:1c2c...	DNS	234	Standard query response 0x7831 A browser.events.
151	17.093861	fe80::1	fe80::1c11:3d:1c2c...	DNS	193	Standard query response 0x7bfa HTTPS onedscolprc
152	17.108262	fe80::1	fe80::1c11:3d:1c2c...	DNS	193	Standard query response 0xcb9d AAAA onedscolprde
231	19.039082	fe80::1c11:3d:1c2c...	fe80::1	DNS	113	Standard query 0xeab3 HTTPS api-safari-aaps1a.sm
232	19.039529	fe80::1c11:3d:1c2c...	fe80::1	DNS	113	Standard query 0x8a7d AAAA api-safari-aaps1a.smc
233	19.039829	fe80::1c11:3d:1c2c...	fe80::1	DNS	113	Standard query 0x680e A api-safari-aaps1a.smoot.

Figure 2: DNS Requests and Responses Captured Using Wireshark

Step 5 – HTTPS Analysis

HTTPS Traffic Analysis: HTTPS is the secure version of HTTP and commonly uses TCP port 443. It uses encryption through TLS to protect data exchanged between a client and a web server.

To identify HTTPS-related traffic, the Wireshark display filter `tcp.port == 443` was applied. Packets using TCP port 443 and TLS traffic were displayed.



The image shows a Wireshark packet capture window with the display filter set to 'tcp.port == 443'. The packet list shows several HTTPS/TLS packets. The packet details pane for packet 1 is expanded, showing the Ethernet II, Internet Protocol Version 6, and Transmission Control Protocol headers.

No.	Time	Source	Destination	Protocol	Length	Info
21855	58.573587	2402:26f7:618:ace8...	2401:4900:8852:c7e...	UDP	117	443 - 59868 Len=55
21856	58.574434	2401:4900:8852:c7e...	2402:26f7:618:ace8...	UDP	91	68199 - 443 Len=29
21857	58.577551	2404:6800:4002:819...	2401:4900:8852:c7e...	QUIC	89	Protected Payload (KP0)
21858	58.584841	2404:6800:4002:819...	2401:4900:8852:c7e...	QUIC	98	Protected Payload (KP0)
21859	58.593043	2404:6800:4013:804...	2401:4900:8852:c7e...	QUIC	89	Protected Payload (KP0)
21860	58.597850	2404:6800:4013:804...	2401:4900:8852:c7e...	QUIC	89	Protected Payload (KP0)
21861	58.598277	2404:6800:4013:811...	2401:4900:8852:c7e...	QUIC	89	Protected Payload (KP0)
21862	58.598617	2401:4900:8852:c7e...	2404:6800:4002:819...	QUIC	91	Protected Payload (KP0), DCID=e5ad10660e86f9f0
21863	58.598807	2401:4900:8852:c7e...	2402:26f7:618:ace8...	QUIC	106	59868 - 443 Len=44
21864	58.622008	2401:4900:8852:c7e...	2404:6800:4002:829...	QUIC	91	Protected Payload (KP0), DCID=ec5ddf94c189ff5a
21865	58.622454	2400:1901:0:5e8a:1...	2401:4900:8852:c7e...	QUIC	89	Protected Payload (KP0)
21866	58.645146	2404:6800:4002:819...	2401:4900:8852:c7e...	QUIC	98	Protected Payload (KP0)
21867	58.650571	2404:6800:4002:829...	2401:4900:8852:c7e...	QUIC	91	Protected Payload (KP0)

Figure 3: Shows HTTPS/TLS Traffic Captured Using Wireshark

Step 6 — Source & Destination IP

Source and Destination IP Address: Every network packet contains addressing information that identifies the source and destination of the communication. In Wireshark, these addresses can be viewed in the Source and Destination columns.

A packet was selected from the capture, and its Internet Protocol information was examined in the packet's details section. The source address represents the device that sent the packet, while the destination address represents the device that received it.

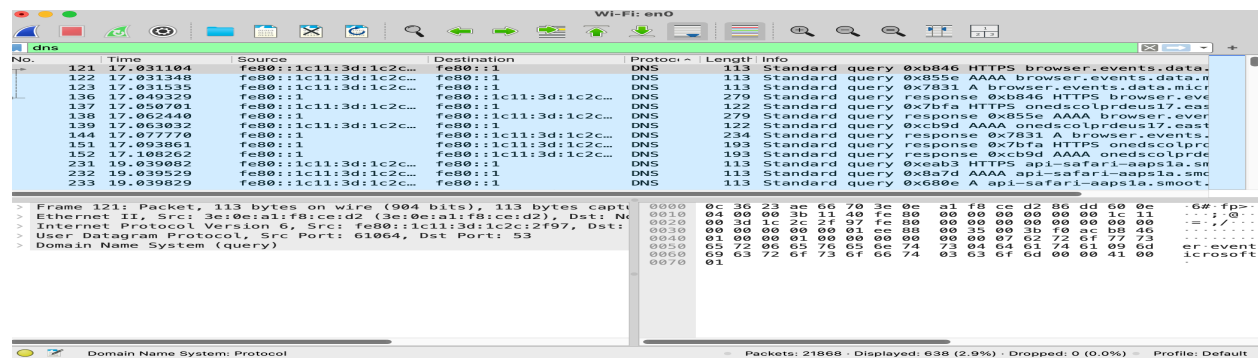


Figure 4: Source and Destination IP Addresses

TASK 5 — NETWORK SCANNING

Step 1 — We must check whether Nmap is installed: Nmap is a network scanning tool used to discover hosts, identify open ports, and detect services running on a networked system. Before performing the scan, I verified that Nmap was installed successfully on my Mac using the Nmap --version command.

```
amaanfayaz — zsh — 120x30
Last login: Thu Sep 10 20:44:12 on ttys000
amaanfayaz@Amaans-MacBook-Air ~ % nmap --version
Nmap version 7.991 ( https://nmap.org )
Platform: x86_64-apple-darwin21.6.0
Compiled with: nmap-liblua-5.4.8 openssl-3.0.21 nmap-libssh2-1.11.1_NMAP1 libz-1.3.2 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: kqueue poll select
amaanfayaz@Amaans-MacBook-Air ~ %
```

Step 2 — Perform a Basic Nmap Scan: I performed a basic Nmap scan on the authorized training target scanme.nmap.org using the command below: **Nmap scanme.nmap.org**

```
amaanfayaz@Amaans-MacBook-Air ~ % nmap scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-10 21:59 +0530
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.30s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo
31337/tcp open  Elite
Nmap done: 1 IP address (1 host up) scanned in 94.80 seconds
amaanfayaz@Amaans-MacBook-Air ~ %
```

Figure 5: Basic Nmap Scan of scanme.nmap.org

Step 3 — Identify Open Ports and Services

The scan showed that the target host was reachable. The open TCP ports identified were 22 (SSH), 80 (HTTP), 9929 (Nping-Echo), and 31337 (Elite). Ports 135, 139, and 445 were shown as filtered.

Step 4 — Service Version Detection

To identify the services and their versions, I performed a service detection scan using the following command: **Nmap -sV scanme.nmap.org**

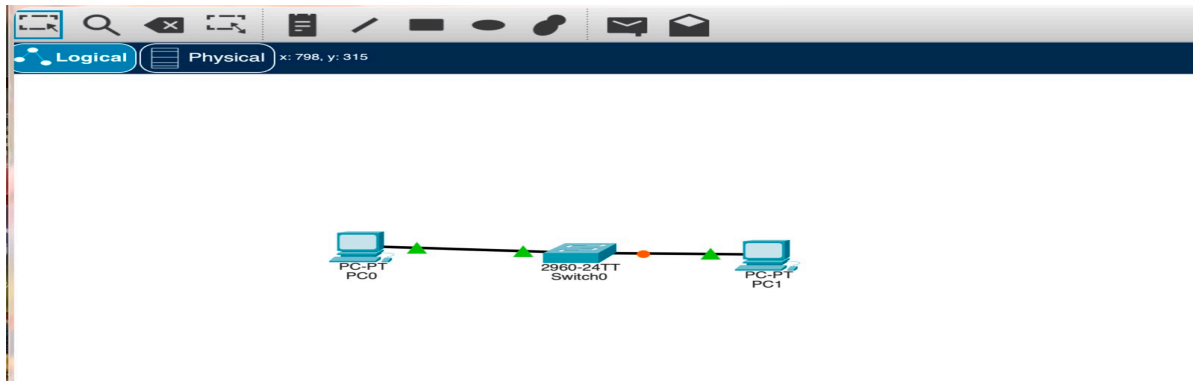
```
amaanfayaz@Amaans-MacBook-Air ~ % nmap -sV scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-10 22:17 +0530
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.85 seconds
amaanfayaz@Amaans-MacBook-Air ~ %
```

Figure 6: Nmap Service and Version Detection

Task 6 — Cisco Packet Tracer

Step 1 — Objective: The objective of this practical is to create a simple local area network using two PCs and a switch, and to verify connectivity between the devices using the ping command.

Step 2 — Network Setup: This step shows Two PCs were connected to a Cisco 2960-24TT switch using Copper Straight-Through cables. PC0 was configured with IP address 192.168.1.10, while PC1 was configured with IP address 192.168.1.11. Both PCs used the subnet mask 255.255.255.0.



This figure shows PC0 → Switch0 → PC1

Step 3 — Connectivity Test: Connectivity was tested from PC0 by using the command ping 192.168.1.11. The test received 4 replies with 0% packet loss, confirming successful communication between the two PCs through the switch.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>192.168.1.11
Invalid Command.
C:\>192.168.1.11
Invalid Command.
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=16ms TTL=128
Reply from 192.168.1.11: bytes=32 time=10ms TTL=128
Reply from 192.168.1.11: bytes=32 time=10ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 9ms
C:\>
```

This fig shows successful Ping Between PC0 and PC1

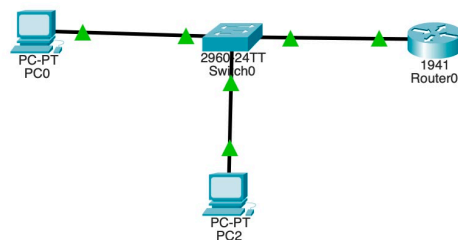
TASK 7 — MITM CONCEPT SAFE SIMULATION

Objective: The objective of this practical is to understand the concept of a Man-in-the-Middle (MITM) attack through a safe network simulation. The activity demonstrates the position of an attacker between communicating devices, the possible risks of MITM attacks, and methods that can be used to prevent them.

Step 2 — MITM concept: A Man-in-the-Middle (MITM) attack occurs when an attacker positions themselves between two communicating devices and attempts to observe or manipulate their communication. For example, on an unsecured public Wi-Fi network, an attacker may try to position themselves between a user's device and the network gateway. This can potentially expose sensitive information such as login credentials, session information, or private communication.

Step 3 — Network Setup: A small network was created in Cisco Packet Tracer to demonstrate the MITM concept safely. The topology consists of PC0 as the victim device, PC2 as the attacker device, Switch0 as the connecting network device, and Router0 as the network gateway. PC0 was assigned to the IP address 192.168.1.10; PC2 was assigned 192.168.1.20, and Router0 was assigned 192.168.1.1. Both PCs use the subnet mask 255.255.255.0.

Step 4 — Normal Connectivity Test: Connectivity was tested from PC0 to PC2 using command ping 192.168.1.20. The test received 4 replies with 0% packet loss, confirming successful communication between the two devices through Switch0.



```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=12ms TTL=128
Reply from 192.168.1.20: bytes=32 time=12ms TTL=128
Reply from 192.168.1.20: bytes=32 time=10ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 8ms

C:\>
```

Now Step 5 — Attacker Position: PC2 represents the attacker in the simulated MITM scenario. Since PC2 is connected to the same Switch0 as the victim PC0, it is positioned on the same local network. In a real MITM attack, an attacker may attempt to place themselves between a victim and the network gateway to observe or manipulate communication. This practical only demonstrates the network position concept and does not perform any real interception.

Step 6 — Public Wi-Fi Example: A common real-world example of a MITM attack can occur on an unsecured or poorly secured public Wi-Fi network. An attacker connected to the same network may attempt to intercept communication between a user's device and the network gateway. This can put sensitive information such as login credentials, session data, and private communication at risk. Using secure HTTPS websites, avoiding unknown Wi-Fi networks, using a trusted VPN, and enabling multi-factor authentication can help reduce the risk.

Step 7 — Risks of MITM Attack: A successful MITM attack can create several security risks. An attacker may capture sensitive information, including usernames, passwords, session information, and private messages. The attacker may also modify communication between the victim and the intended destination. Other risks include identity theft, financial loss, privacy violations, and unauthorized access to accounts or systems. The severity of the impact depends on the type of information exposed and the security controls being used.

Step 8 — Prevention Methods: MITM attacks can be reduced by using strong network security practices. Users should prefer secure HTTPS websites and avoid connecting to unknown or unsecured public Wi-Fi networks. A trusted VPN can provide additional protection on untrusted networks. Multi-factor authentication (MFA) should be enabled for important accounts. Network administrators can also use encryption, secure Wi-Fi protocols, monitoring, firewalls, and proper access controls to detect and prevent unauthorized interception.

Conclusion: This practical provided a clear understanding of the Man-in-the-Middle (MITM) attack concept through a safe Cisco Packet Tracer simulation. The network topology demonstrated how an attacker device can exist on the same local network as a victim device. The successful connectivity test confirmed communication between the devices, while the theoretical discussion highlighted the risks associated with unauthorized interception of network traffic. No real MITM attack or data interception was performed. Overall, this activity improved my understanding of network security, common MITM risks, and the importance of using secure networks, encryption, HTTPS, VPNs, and multi-factor authentication to protect sensitive information.